

# HawkEye Keylogger

## Advanced Deep Research Intelligence Report

### 1. Executive Summary

HawkEye Keylogger is a Windows-based information-stealing malware family used primarily for credential theft, surveillance, and data exfiltration. Originally marketed as a commercial monitoring tool, it evolved into widely abused malware distributed through phishing campaigns and underground crimeware channels. Its operational model and repeated rebranding have made it a persistent threat in the malware ecosystem.

Modern HawkEye variants are capable of keylogging, browser credential theft, clipboard monitoring, screenshot capture, process injection, and exfiltration over SMTP, FTP, HTTP, and in some newer environments, Telegram-based channels. The malware is especially dangerous because it combines persistence, stealth, and broad data collection with relatively low attacker skill requirements.

### 2. Malware Classification

| Field             | Details                                              |
|-------------------|------------------------------------------------------|
| Malware Name      | HawkEye Keylogger                                    |
| Malware Type      | Keylogger / Infostealer / Remote Access Malware      |
| Platform          | Windows                                              |
| Delivery Model    | Malware-as-a-Service / Commercial crimeware          |
| Primary Objective | Credential theft and surveillance                    |
| Common Delivery   | Phishing emails, malicious attachments, and archives |
| Threat Severity   | High                                                 |

| Field            | Details                 |
|------------------|-------------------------|
| Malware Family   | HawkEye Keylogger       |
| Malware Category | Keylogger / Infostealer |
| Operating System | Windows                 |

|                      |                                        |
|----------------------|----------------------------------------|
| Threat Level         | High                                   |
| Discovery Period     | Early 2010s                            |
| Current Status       | Active and Evolving                    |
| Primary Targets      | Business Users and Organizations       |
| Delivery Method      | Phishing Emails, Malicious Attachments |
| Main Objective       | Credential Theft and Surveillance      |
| Related Malware      | PredatorPain, Agent Tesla              |
| Associated Campaigns | HawkEye Reborn Campaigns               |

### 3. Discovery and Timeline

HawkEye has been active for many years and has appeared in multiple campaign waves. Early samples emerged in the early 2010s, followed by broader spam-driven credential theft campaigns. In later years, the malware evolved with improved stealth, more modular execution chains, and stronger anti-analysis features.

| Year      | Event                                                                          |
|-----------|--------------------------------------------------------------------------------|
| 2013      | Early HawkEye samples observed.                                                |
| 2014      | Spam-based credential theft campaigns increased.                               |
| 2016      | Global phishing distribution expanded.                                         |
| 2018      | HawkEye Reborn v8 introduced stronger evasion and theft capabilities.          |
| 2019      | Malicious archives and Office-based delivery became more common.               |
| 2021      | Loader obfuscation and targeted credential collection increased.               |
| 2022      | Telegram-style exfiltration patterns appeared in related malware ecosystems.   |
| 2024–2026 | Continued MaaS-style circulation in underground forums and phishing campaigns. |

## 4. Initial Access and Delivery

HawkEye is most commonly delivered through phishing emails that impersonate legitimate business communication. Typical lures include invoices, purchase orders, shipment notifications, airline tickets, and HR-related documents.

Common delivery formats include:

- .docm
- .xlsm
- .exe
- .7z
- .rar
- .vbs

These delivery methods align with its use in socially engineered campaigns designed to trick users into enabling macros, opening archives, or launching disguised executables.

## 5. Malware Architecture

HawkEye typically includes several functional components that support its collection and exfiltration workflow.

| Module               | Purpose                                           |
|----------------------|---------------------------------------------------|
| Keylogger Engine     | Captures keyboard input.                          |
| Clipboard Monitor    | Tracks copied data.                               |
| Screenshot Module    | Captures desktop screenshots.                     |
| Credential Stealer   | Extracts browser and email credentials.           |
| Loader Module        | Decrypts and launches payloads.                   |
| Persistence Module   | Maintains long-term access.                       |
| Exfiltration Engine  | Sends stolen data to attacker infrastructure.     |
| Anti-Analysis Module | Detects sandboxes, debuggers, and security tools. |

## **6. Infection Chain**

The infection process typically follows a predictable sequence:

1. The victim receives a phishing email.
2. The user opens a malicious attachment or archive.
3. A loader executes and decrypts the HawkEye payload.
4. The malware injects itself into legitimate Windows processes.
5. Persistence is established through registry or startup mechanisms.
6. Credentials, screenshots, and system details are collected.
7. The stolen data is transmitted to attacker-controlled infrastructure.

This layered workflow is designed to reduce detection and improve long-term access to the infected system.

## **7. Process Injection and Memory Techniques**

HawkEye Reborn variants have been documented using advanced execution and memory manipulation techniques to evade detection. These include process hollowing, .NET reflection, in-memory unpacking, and native Windows API usage.

Reported target or injected processes include:

- MSBuild.exe
- RegAsm.exe
- VBC.exe

These legitimate binaries are abused to conceal malicious activity and make behavioral detection more difficult. This technique helps the malware blend into normal Windows activity while executing its payload in memory.

## **8. Credential Theft Capabilities**

HawkEye is optimized for credential harvesting and data theft. It is known to target:

- Web browsers such as Chrome, Firefox, Internet Explorer, and Opera.
- Email clients such as Outlook and Thunderbird.
- Communication or collaboration tools.
- FTP clients.
- VPN-related applications.

- In some cases, wallet or payment-related data.

The malware's focus on stored credentials and active session data makes it particularly useful for account compromise, lateral movement, and follow-on fraud.

## **9. Keylogging and Monitoring Behavior**

HawkEye typically collects user input and active session data through multiple monitoring techniques:

- Keyboard hooks.
- Mouse activity monitoring.
- Window title tracking.
- Clipboard interception.
- Active window capture.
- Screenshot collection.

In ATT&CK terms, this behavior maps closely to keylogging and screen capture techniques. It is especially relevant for identifying data theft from high-value users such as finance, operations, and executive personnel.

## **10. Persistence Mechanisms**

HawkEye frequently uses standard Windows persistence methods, including:

- Registry Run keys.
- Startup folder entries.
- Scheduled tasks.
- Service-based persistence.
- Auto-run modifications.

Example persistence location:

- HKCU\Software\Microsoft\Windows\CurrentVersion\Run

These mechanisms help the malware survive reboots and maintain long-term foothold in the environment.

## **11. Defense Evasion**

HawkEye includes several anti-analysis and defense evasion features. Reported behaviors include:

- Obfuscation of code and strings.
- Delayed execution to evade sandbox timeouts.
- Virtual machine detection.
- Debugger checks.
- Security tool monitoring.

It has also been observed identifying tools such as:

- Wireshark.
- Sandboxie.
- Process Hacker.
- Process Explorer.

It may also target security products or related processes such as:

- MsMpEng.exe
- mbam.exe
- avp.exe
- wireshark.exe

This suggests the malware actively checks for analysis environments and attempts to reduce the chance of inspection.

## **12. Host Modification and Hardening of Persistence**

Some HawkEye variants modify the Windows HOSTS file to block access to security vendor domains, interfere with updates, or prevent access to defensive websites. This can delay remediation and make the system harder to clean.

This technique supports the malware's broader objective of prolonging dwell time and reducing the victim's ability to detect or remove it.

## **13. Command and Control Infrastructure**

HawkEye commonly communicates using:

- SMTP.

- FTP.
- HTTP/HTTPS.
- Telegram APIs in newer variants.

SMTP remains one of the most frequently reported exfiltration channels. The malware often uses legitimate or compromised email infrastructure to send stolen logs and credentials, which helps it hide within normal outbound business traffic.

Observed infrastructure patterns include disposable phishing domains, shared hosting providers, dynamic DNS services, low-cost VPS infrastructure, and rapidly rotating SMTP accounts. These operational patterns are consistent with commodity malware ecosystems designed for quick deployment and large-scale credential harvesting rather than long-term stealth persistence.

## 14. Exfiltration Workflow

The malware may collect and transmit the following data:

- Passwords.
- Cookies.
- Clipboard contents.
- Screenshots.
- System information.
- Browser history.
- Active window data.

Common exfiltration destinations include:

- Gmail SMTP.
- Compromised mail servers.
- FTP servers.
- Remote control panels.
- Telegram bots.

This makes HawkEye especially effective for stealing both identity data and operational intelligence.

## 15. MITRE ATT&CK Mapping

| ATT&CK ID | Technique |
|-----------|-----------|
|-----------|-----------|

|           |                                   |
|-----------|-----------------------------------|
| T1056.001 | Keylogging                        |
| T1113     | Screen Capture                    |
| T1027     | Obfuscated Files or Information   |
| T1055     | Process Injection                 |
| T1547     | Boot or Logon Autostart Execution |
| T1082     | System Information Discovery      |
| T1041     | Exfiltration Over C2 Channel      |
| T1071     | Application Layer Protocol        |
| T1105     | Ingress Tool Transfer             |
| T1005     | Data from Local System            |

This mapping should be validated against each sample you analyze, but it provides a strong baseline for intelligence reporting.

## 16. Indicators of Compromise

### Common File Names

- invoice.exe
- payment\_copy.exe
- scan\_receipt.exe
- DHL\_Receipt.exe

### Common Directories

- %APPDATA%
- %TEMP%
- %PROGRAMDATA%

### Suspicious Network Patterns

- Outbound SMTP sessions.
- Unexpected FTP traffic.
- Encoded HTTP POST requests.
- Telegram API communication.



## Host Artifacts

- Registry autoruns.
- Startup-folder persistence.
- Malicious scheduled tasks.
- Modified HOSTS file entries.
- Suspicious dropped files in user-writable directories.

### 16.1 Infrastructure Intelligence

HawkEye infrastructure commonly relies on SMTP relays, disposable domains, shared hosting services, and lightweight web panels used for credential collection and campaign management. Threat actors frequently rotate infrastructure to avoid blacklisting and detection.

Common infrastructure behaviors include:

- Dynamic DNS usage.
- Disposable email accounts.
- Shared VPS hosting.
- SMTP-based credential exfiltration.
- Lightweight HTTP command panels.
- Rapid domain rotation.

This infrastructure model supports fast deployment, scalability, and reduced operational cost for attackers distributing HawkEye campaigns.

## 17. Campaign Themes and Victimology

| Campaign          | Period | Target Sector          | Delivery Method | Assessment               |
|-------------------|--------|------------------------|-----------------|--------------------------|
| HawkEye Reborn v8 | 2018   | Business Organizations | Phishing Emails | Credential theft focused |

| Campaign                   | Period    | Target Sector         | Delivery Method          | Assessment                     |
|----------------------------|-----------|-----------------------|--------------------------|--------------------------------|
| Invoice-themed Campaigns   | 2019–2021 | Finance and Logistics | Malicious Attachments    | Data theft and persistence     |
| HR and Resume Campaigns    | 2021–2024 | Enterprises           | Email Attachments        | Credential harvesting          |
| MaaS Distribution Activity | 2024–2026 | Multi-sector          | Underground distribution | Continued infostealer activity |

| Source                  | Intelligence Type                 | Relevance |
|-------------------------|-----------------------------------|-----------|
| Microsoft Security Blog | Technical Malware Analysis        | High      |
| Malwarebytes            | Detection and Behavioral Analysis | High      |
| Cisco Talos             | Campaign Intelligence             | High      |
| Security Affairs        | Infrastructure Reporting          | Medium    |
| ANY.RUN Sandbox         | Behavioral Analysis               | High      |
| MITRE ATT&CK            | TTP Mapping                       | High      |

**The intelligence assessment in this report is based on publicly available threat intelligence reports, malware analysis research, behavioral observations, and ATT&CK framework mapping.**

HawkEye is frequently distributed using realistic business lures. Common themes include:

- Invoice campaigns.
- Shipping notifications.
- Financial requests.
- HR or resume attachments.
- Travel or airline confirmations.

Victimology has included organizations and users in sectors such as:

- Banking.
- Healthcare.
- Engineering.
- Government.
- Logistics.
- Manufacturing.

The malware is often used against business users because those environments are more likely to process attachments and rely on email-based workflows.

## **18. Detection Engineering Notes**

A strong detection strategy should combine endpoint, email, and network telemetry. Focus areas include:

- Office spawning scripting engines.
- Unusual SMTP from user workstations.
- Registry autorun creation.
- Screenshot-related API usage.
- Browser credential file access.
- Process injection into legitimate binaries.

### **Example Sigma Logic**

text

title: HawkEye Suspicious SMTP Exfiltration

logsource:

category: network\_connection

detection:

selection:

DestinationPort: 587

condition: selection

level: high

This should be treated as a starting point, not a complete detection rule.

## 19. YARA Starting Point

A YARA rule should be written carefully and validated against known samples. The goal is to detect stable traits such as strings, embedded configuration markers, or structural patterns rather than overly generic terms.

Example starter structure:

text

rule HawkEye\_Keylogger\_Advanced

{

meta:

description = "Detect HawkEye malware behavior"

author = "Threat Research"

strings:

\$smtp1 = "smtp.gmail.com"

\$smtp2 = "mail.smtp"

\$host1 = "hosts"

\$clip = "clipboard"

\$key = "keylogger"

condition:

3 of them

}

This is only a research skeleton and should be refined with sample-specific analysis.

## 20. Incident Response Guidance

### Immediate Actions

- Isolate infected endpoints.
- Block outbound SMTP abuse.
- Reset exposed credentials.

- Remove persistence artifacts.
- Review outbound logs.
- Scan for secondary payloads.

### **Long-Term Controls**

- Enforce MFA.
- Deploy EDR with behavioral detection.
- Harden email security.
- Restrict macros.
- Monitor DNS and outbound traffic.
- Enable PowerShell logging.

These steps reduce both containment time and the chance of repeated compromise.

## **21. Threat Assessment**

HawkEye remains a high-risk credential theft threat because it is inexpensive, widely distributed, and effective against ordinary enterprise email workflows. It combines persistence, stealth, and exfiltration methods that are simple enough for low-skill operators yet damaging enough for real-world compromise.

Its MaaS-style ecosystem lowers the entry barrier for attackers and supports repeated reuse across different campaigns, which is why it remains relevant in contemporary threat intelligence.

## **22. Final Intelligence Summary**

HawkEye Keylogger is more than a basic keylogger. It is a mature infostealer family with phishing-based delivery, credential harvesting, process injection, persistence, anti-analysis features, and flexible exfiltration methods.

Its most important research value lies in how it illustrates the evolution of commodity crimeware into a modular, evasive, and commercially distributed threat. For defenders, that means detection must go beyond hashes and focus on behavior, infrastructure, and campaign clustering.

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